

Conformal Calibration of Open-Vocabulary Semantic Maps under Label and Pose Uncertainty

Sergey Shchetinkin

ITMO University, Saint Petersburg, Russia

Abstract—A robot that avoids “the plant” on an open-vocabulary semantic map faces two errors at once: the detector misses, truncates or mislabels objects, and the drifting pose estimate draws them in the wrong place. Distribution-free guarantees for planning on perception maps usually calibrate the perception error at a known pose. We calibrate one per-scene quantity, the *miss distance*: how far the true footprints of a class reach beyond the class region on the map built with estimated poses, measured in the planner frame. Its split-conformal quantile is a keep-out margin that certifies every path, whichever planner produced it, and abstains when the margin would be uninformative. On 21 OSMa-Bench scenes, margins calibrated on labels or on pose alone lose coverage under the other error, and the sum of the two over-pays by up to a quarter. Label-set calibration degenerates to keeping out every occupied cell, because the detector truncates the edge of every object. With few calibration scenes the price of the guarantee is set by one or two worst scenes: a single missed plant makes its class abstain in a quarter to two thirds of the splits. Fusing three detectors by the maximum nearly removes abstention, while verifying detections with CLIP brings it back: misses set the margin, so recall matters more than precision. Calibrating the expected share of unsafe missions instead of the map keeps most of the uncalibrated planner’s successes with at most 0.6% unsafe missions on average instead of 7–8%. Plans certified at query time stay safe from the avoid classes when executed under continued drift of up to 7 cm. With visual odometry there is no useful guarantee; loop-closure SLAM cuts the median pose error threefold, but the scenes where it finds no loop set the margin.

I. INTRODUCTION

Open-vocabulary semantic maps [1] let a robot follow instructions such as “keep away from the bike”. The map has two unreliable inputs. A detector or segmenter supplies the labels, and it misses objects, covers them only in part, or confuses classes. The robot’s pose estimate places the observations, and it drifts. Planners usually take the map at face value and keep a fixed clearance from the regions of avoid classes.

How unsafe this is usually goes unmeasured. OSMa-Bench [2] shows that open-vocabulary maps depend noticeably on lighting (on ReplicaCAD the f-mIoU of ConceptGraphs drops from 28.00 to 14.29 under dynamic lights), but it scores segmentation and scene-graph answers rather than a robot moving through the map. A recent survey of 3D scene graphs [3] lists the link between map errors and robot behaviour as an open problem.

Conformal prediction (CP) [4] gives a planner guarantees without assumptions on the error distribution. Sundarsingh et

al. [5] calibrate per-cell label sets of a semantic grid, and Perceive With Confidence [6] calibrates how much to inflate an occupancy map; both assume a known robot pose. Separately calibrated modules need not form a calibrated system [7].

We ask what guarantee a planner gets when label error and pose error act together, and how to calibrate it. Contributions:

- the *miss distance*, a per-class version of the inflation score of [6] measured in the planner frame on a map built with estimated poses; its conformal quantile is a margin that covers both errors for any planner and abstains when it would be uninformative;
- an analysis of joint versus separate calibration: the sum of two separately calibrated margins guarantees only $1 - 2\alpha$, under a premise that geometry does not imply, and we construct a case where it under-covers;
- an evaluation on 21 OSMa-Bench scenes with five levels of synthetic drift, real visual odometry and loop-closure SLAM, MobileSAM masks and a transfer to HM3D, which reports the price of the guarantee (margin, abstention, solved tasks) next to coverage and compares the method with label-set calibration and with a margin without semantics;
- a study of what lowers the price: detector ensembles and a CLIP-verified cascade, a mission-level guarantee by conformal risk control, and execution under continued drift.

II. RELATED WORK

Open-vocabulary mapping. ConceptGraphs [1] fuses class-agnostic masks into an object graph with CLIP features. OSMa-Bench [2] and OSMa-Bench++ [8] evaluate such mappers by 3D segmentation and scene-graph question answering under lighting changes, without a motion task. Naive Bayesian fusion makes semantic maps over-confident [9]; we fuse by averaging. Tchuiev and Indelman [10] keep a joint Bayesian belief over robot poses, object poses and classes and plan in belief space; the approach is model-based rather than distribution-free.

Conformal guarantees for planning. Conformal prediction (CP) is used in planning, for example, for trajectory prediction of other agents [11]. Yang et al. [12] calibrate perception-based state estimation error and build control barrier functions robust to it, without a map or semantics. Shin et al. [13] conformalize the whole distance field to moving obstacles and obtain a constraint that certifies any MPC trajectory; they name folding perception and state-estimation error into

the same pipeline as future work. SCORE [14] defines the score on a signed distance field to obtain non-convex keep-out regions. Learnable CP [15] learns a context-aware score for planner clearance and includes synthetic localization noise in its ablations, and planner-agnostic constraint tightening [16] bounds the deviation between nominal and true dynamics. Closest to us, Sundarsingh et al. [5] calibrate label sets of grid cells with one score per environment, the worst cell observed along sampled paths, and plan with the most conservative label of each cell. With a known pose, five classes (including “free”), simulated 70×25 m sites and a calibration-conditional variant ($\delta = 0.1$), they report 93.36% correct maps and 98.36% mission success at a 95% target. Perceive With Confidence (PwC) [6] calibrates the smallest inflation of an occupancy map that covers all obstacles; our miss distance is a per-class version of that score, measured in the planner frame on a map built with estimated poses.

Composition and pose error. In a synthetic Gaussian example, two modules calibrated at 95% and composed under an independence assumption give 99.98% at error correlation -0.9 and 86.46% at $+0.9$, and estimating the joint covariance restores the target [7]. Calafiore [17] adds blockwise certificates through a union bound. Mahajan et al. [18] observe that local conformal guarantees of occupancy maps degrade because correlation along a trajectory breaks exchangeability, and recover coverage by fusing maps of several robots; weighting calibration points bounds the coverage loss without exchangeability [19]. Pose error in semantic memory is real and poorly calibrated: on a real robot the nominal 95% ellipsoid of object position derived from the pose graph covers 33% of re-detections [20]. A citation and keyword search (October 2026) found no work that calibrates open-vocabulary label error and robot pose error jointly with a planner in the loop; the search is not exhaustive.

III. PROBLEM FORMULATION

A scene Ω is drawn from an unknown distribution $\mathcal{D}$. It contains objects of avoid classes $\mathcal{K}$; object e has class k_e and floor footprint $F_e \subset \mathbb{R}^2$, and $F_k = \bigcup_{e: k_e=k} F_e$. The robot moves along true poses $T_1, \dots, T_Q \in SE(2)$ and receives a task at time Q . Its estimator returns $\hat{T}_t = E_t T_t$ with accumulated planar error $E_t \in SE(2)$. A grid with cell size Δ is fused from all observations placed with the *estimated* poses; cell j stores the fraction $p_j(k)$ of its observations that support class k . The region of class k at threshold λ is the set of cell centres

$$A_\lambda(k) = \{c_j : \text{cell } j \text{ occupied, } p_j(k) \geq \lambda\}. \quad (1)$$

The robot believes its pose is $\hat{T}_Q$, so in the planner frame the true footprint is $\hat{F}_k = E_Q F_k$. A point observed at time t is drawn at $E_t x = (E_t E_Q^{-1}) E_Q x$, i.e., displaced from that truth by the relative error $E_t E_Q^{-1}$. A motion G common to the whole trajectory (estimates $G E_t T_t$) moves the map and the robot alike and changes nothing; only relative errors matter.

A path is a sequence of robot-centre positions; it is safe for class k if every point x on it has $\text{dist}(x, \hat{F}_k) \geq d$, with $d = 0.5$ m.

Assumption 1 (execution). The robot follows the path in the planner frame, and the pose error does not change during execution.

The guarantee is thus about the plan made at time Q , not about closed-loop execution. Given n calibration scenes with known ground truth, we seek per-class margins $\hat{r}_k \in [0, \infty]$, as small as possible, such that for a new scene

$$\mathbb{P} \left(\begin{array}{l} \text{every path with } \text{dist}(\gamma, A_\lambda(k)) > d + \hat{r}_k \\ \text{is safe for class } k \end{array} \right) \geq 1 - \alpha. \quad (2)$$

The probability is over the calibration scenes, the new scene and the error realizations. The value $\hat{r}_k = \infty$ (*abstention*) satisfies (2) trivially by admitting no path.

Assumption 2 (exchangeability). The pairs (Ω_i, ξ_i) , $i = 1, \dots, n+1$, of a scene and the realization of perception and pose errors in it are exchangeable, and the detector, the mapping, λ and $R_{\max}$ are fixed before calibration.

IV. METHOD

Definition 1 (miss distance). Let $\hat{F}_k$ be the centres of the cells hit by points of the true objects of class k moved to the planner frame (cells beyond the map edge are kept). For a scene and class k ,

$$s_k = \min\{R_{\max}, h(\hat{F}_k, A_\lambda(k))\}, \quad h(X, Y) = \max_{x \in X} \min_{y \in Y} \|x - y\|, \quad (3)$$

the directed Hausdorff distance; $s_k = 0$ if the scene has no object of class k and $s_k = R_{\max}$ if the region is empty.

A missed or mislabeled object gives a large score, a truncated object the length of the missing part, and a map shift about its size, with a lever-arm effect for rotations. We compute one score per scene: cells, frames and objects of a scene are strongly dependent and are not exchangeable units.

Calibration. Let $s^{(1)} \leq \dots \leq s^{(n)}$ be the sorted calibration scores of class k and $m = \lceil (n+1)(1-\alpha) \rceil$. The margin is $\hat{r}_k = s^{(m)}$ if $m \leq n$ and $s^{(m)} < R_{\max}$, and $\hat{r}_k = \infty$ otherwise. With $n = 13$ and $\alpha = 0.1$, $m = 13$: the margin is the largest calibration score. Calibrating per class keeps one unreliable class from inflating the others.

Proposition 1. *Under Assumptions 1 and 2, $\mathbb{P}(\hat{r}_k = \infty \text{ or } s_k(\Omega_{n+1}) \leq \hat{r}_k) \geq m/(n+1) \geq 1-\alpha$, and $\hat{r}_k$ satisfies (2) for paths through cell centres.*

Proof. By exchangeability the rank of $s_k(\Omega_{n+1})$ among the $n+1$ scores is uniform, and ties only help, so $\mathbb{P}(s_k(\Omega_{n+1}) \leq s^{(m)}) \geq m/(n+1)$ [4]; under abstention the event holds. If $\hat{r}_k < \infty$ and $s_k \leq \hat{r}_k$, every $y \in \hat{F}_k$ has some $a \in A_\lambda(k)$ with $\|y - a\| \leq \hat{r}_k$. For a path point x with $\text{dist}(x, A_\lambda(k)) > d + \hat{r}_k$, the triangle inequality gives $\|x - y\| \geq \|x - a\| - \|a - y\| > d$. $\square$

The certificate does not depend on the planner, which only decides whether a path exists in the narrowed free space. For continuous footprints and motion between cell centres the distance can fall short of d by at most $\sqrt{2}\Delta \approx 7$ cm at the true pose and $3\Delta/\sqrt{2} \approx 11$ cm under drift, where moving the

footprint cells to the planner frame adds another $\Delta/\sqrt{2}$. The guarantee is per class, so with mission classes $\mathcal{H}_m$ the union bound gives $1 - |\mathcal{H}_m| \alpha$ for all of them. It is marginal over calibration sets: for $n = 13$ the coverage of a fixed calibration set follows Beta(13, 1), with mean 0.93 and 10th percentile 0.84.

Compared calibrations. *Uncalibrated:* region $A_{\lambda_0}(k)$, margin 0. *Labels only:* the score (3) of the detector map at true poses, margin $\hat{r}_k^{\text{lab}}$. *Pose only:* the score of a ground-truth-label map built with drifted poses, $\hat{r}_k^{\text{pose}}$. *Sum:* $\hat{r}_k^{\text{lab}} + \hat{r}_k^{\text{pose}}$, both at level α . *Joint (ours):* the score of the detector map built with drifted poses. These five share the region $A_{\lambda_0}(k)$, $\lambda_0 = 0.05$, chosen on a development scene. *Geometry only:* every occupied cell may be the class ($\lambda = 0$), with a margin calibrated on the drifted map; this is the PwC-style certificate without semantics. *Label sets* [5]: the score is the worst $1 - p_j(k)$ over true footprint cells (∞ if a cell is unoccupied), $\hat{\eta}$ is its conformal quantile, and cells with $p_j(k) \geq 1 - \hat{\eta}$ are avoided, with no metric margin. Their score takes the maximum over all observed cells and classes; restricting it to footprint cells can only lower it. Arms without a margin count as covering a miss of up to one cell, which favours them.

Proposition 2. *If $s_k \leq s_k^{\text{lab}} + s_k^{\text{pose}}$ for the scores of the new scene, then $\mathbb{P}(s_k \leq \hat{r}_k^{\text{lab}} + \hat{r}_k^{\text{pose}}) \geq 1 - 2\alpha$.*

Proof. Each margin fails to cover its own score with probability at most α ; by the union bound both cover with probability at least $1 - 2\alpha$, and then the premise gives the claim. $\square$

The premise does not follow from geometry. The triangle inequality gives $s_k \leq s_k^{\text{lab}} + h(E_Q A^0, A)$, where A^0 is the detector region at true poses: the second term is the displacement of the *predicted* region, while the pose-only score measures a ground-truth-label map, as an engineer tuning localization separately from perception would. In our data the premise fails in a fraction of realizations (Sec. VI-E). A budget of $\alpha/2$ per module [17] restores $1 - \alpha$ but needs $n \geq 2/\alpha - 1 = 19$ at $\alpha = 0.1$. The joint margin does not depend on the unknown error dependence.

Missions. A robot of radius 0.2 m must reach a goal. The keep-out zone of class k is the set of cell centres within $d + \hat{r}_k$ of its region; a class that abstains falls back to the geometry-only certificate. Dijkstra’s algorithm on the 8-connected grid is complete on the grid graph, so “no path” means that none exists. A path is *unsafe* if one of its cell centres is closer than d to a cell of a true avoid object in the planner frame. It *collides* if it comes closer than $0.2\text{ m} - \Delta$ to a true obstacle (the obstacle map is not calibrated, and a one-cell shift is not counted as a crash). A mission *succeeds* if a path is found, is safe and does not collide. A shortest path tests the certificate along one line only, so we also record the worst case over all planners: whether a cell closer than d to a true object is reachable from the start without entering the keep-out zone. The same planner on the true map is an oracle.

Hypotheses. With $\text{Cov}(r)$ the left side of (2): a labels-only margin has $\text{Cov} < 1 - \alpha$ once drift is comparable to the label

error (H1); a pose-only margin has $\text{Cov} < 1 - \alpha$ while drift is smaller than the label error (H2); the joint margin is no larger than the sum, $\hat{r}_k^{\text{joint}} \leq \hat{r}_k^{\text{lab}} + \hat{r}_k^{\text{pose}}$ (H3); map mIoU is a poor predictor of mission safety (H4). The coverage of the joint margin is not a hypothesis: it follows from Proposition 1, and measuring it checks the implementation.

V. EXPERIMENTAL SETUP

Data and perception. We use 22 ReplicaCAD scenes of OSMa-Bench [2], [21] with RGB-D, ground-truth semantics and poses at 640×480 , every 8th frame. All design choices were made on apt_0, which is never used for calibration or testing. YOLO-World v2-s [22] runs with the 98 object classes of the dataset as prompts; inside each box we keep pixels within 0.25 m of the median box depth. Maps are 5 cm grids, and points 0.1–1.8 m above the floor are obstacles. The avoid classes are indoor plant, bike and TV stand; a true footprint is a connected component of at least 8 occupied cells with at least two points of the class.

Pose error. Synthetic drift replays every relative motion of the true trajectory with planar noise proportional to the distance travelled and the angle turned. Besides the true pose (L0), five levels are set on the development scene, L1–L3 by absolute trajectory error (ATE) and L4–L5 by endpoint drift. On the evaluation scenes the median ATE is 0.4 cm (L1), 2 cm (L2), 7 cm (L3), 45 cm (L4) and 2.7 m (L5); we draw 10 realizations per level. As real pose error we run frame-to-frame RGB-D odometry of Open3D [23] (hybrid photometric and geometric term) on every 2nd frame with a planar constraint and no loop closure, without tuning it on ground truth.

Coverage protocol. $\alpha = 0.1$ unless stated, $R_{\text{max}} = 3$ m. The 21 evaluation scenes are split 200 times at random into 13 calibration and 8 test scenes (all splits are enumerated when there are at most 2000). A calibration scene contributes one random drift realization, a test scene all of them. Coverage is the fraction of test realizations whose score is within the margin, averaged over splits, with abstention counted as covered as in Proposition 1; we also report the share of splits with a certificate and the coverage among them. Because splits are drawn from one fixed set of scenes, the scenes are exchangeable by construction and the joint coverage is at least $1 - \alpha$ on average for any data (up to the random choice of realizations). It is a check of the implementation; the margin, the abstention rate and the behaviour of the arms without a guarantee are what the experiment measures.

Tasks. Start and goal are at least 1.5 m from avoid objects and 3 m apart. *Hard* tasks are selected so that the shortest obstacle-only path passes within 0.5 m of a plant or bike (233 tasks in 15 scenes); *random* tasks are not selected (20 per scene, 420 in total). Margins for a scene are calibrated on the other 20 scenes ($m = 19$). The mission avoid classes are plant and bike. Both the task family and this choice were made after first runs: an earlier task family with goals near the objects was degenerate, and the TV stand was removed after it turned out to abstain. The coverage guarantee does not depend on these choices, but mission rates should be read with them in mind.

TABLE I

COVERAGE AND MEDIAN MARGIN (M), $1 - \alpha = 0.9$, 200 SPLITS 13/8. MEDIAN ATE: L2 2 CM, L4 45 CM. ABSTENTION COUNTS AS COVERED; [†] THE ARM ABSTAINS IN AT LEAST HALF OF THE SPLITS, THE MARGIN OF THE OTHERS IS SHOWN. THE GEOMETRY-ONLY MARGIN IS MEASURED FROM ALL OCCUPIED CELLS.

calibration	bike				indoor plant			
	L0	L2	L4	$\hat{r}$ L4	L0	L2	L4	$\hat{r}$ L4
uncalibrated	0.00	0.00	0.01	0	0.00	0.00	0.00	0
label sets [5]	1.00	0.73	0.18	0	1.00	0.41	0.03	0
geometry only	1.00	0.96	0.93	0.65	1.00	0.95	0.93	0.71
labels only	0.93	0.93	0.76	0.75	0.93	0.96	0.86	0.85 [†]
pose only	0.00	0.06	0.82	0.85	0.00	0.01	0.85	1.37
sum	0.93	0.98	0.97	1.61	0.93	0.98	0.95	2.06 [†]
joint (ours)	0.93	0.94	0.92	1.23	0.93	0.94	0.94	1.54 [†]

VI. RESULTS

A. Coverage under pose drift

The uncalibrated map almost never contains the whole true footprint, because the detector truncates object edges (Table I). Label sets reach 1.00 at the true pose only because their threshold $1 - \hat{\eta}$ is zero: in all 21 scenes every class has an occupied footprint cell with no observation of that class (a truncated edge), so the score is 1 and every occupied cell is avoided, exactly the zone of the geometry-only arm at L0. With no metric slack, 2 cm of drift drops their bike coverage to 0.73. The labels-only margin holds while drift stays below the label error and falls to 0.76 for the bike at 45 cm, with a 90 % scene-bootstrap interval of 0.67–0.82 (H1); for the plant the drop is smaller (0.86) and its interval, 0.59–0.90, only touches the target. The pose-only margin cannot see detector errors and stays at or below 0.27 up to 7 cm (H2). The sum over-pays: at 45 cm the joint margin is 23–25 % smaller (1.23 vs. 1.61 m for the bike, 1.54 vs. 2.06 m for the plant), and 6–25 % smaller at 0.4–7 cm (H3).

For plant and bike the joint margin holds 0.92–0.94 at L0–L4, as it must by construction. Its price and abstention are what matter. The bike is certified at every level up to L4 and its margin grows from 0.75 to 1.23 m. The plant is missed entirely in one of the 21 scenes; with 13 calibration scenes the margin is the maximum, so at L0 the class abstains whenever that scene is in the calibration set (63 % of the splits, about 13/21); under drift that scene’s miss is sometimes below the cap and abstention is 24–59 %. When a certificate is issued, the worst scenes tend to be in the test set and the coverage among certified splits is 0.81–0.92, below the target, which a marginal guarantee allows. The TV stand, confidently labeled as a bench, rug or bed in three scenes, abstains in 75–95 % of the splits. Both mission classes are covered at once in 85–87 % of test realizations, consistent with the union bound $1 - 2\alpha$.

B. Missions

On hard tasks the uncalibrated planner almost always finds a path, but 54–58 % of its paths are unsafe at L0–L3: it grazes the truncated edge of the object (Fig. 1, Table II). The joint, sum and labels-only margins have 0–2 unsafe paths out of 74–384, and in the worst case over all planners a dangerous cell

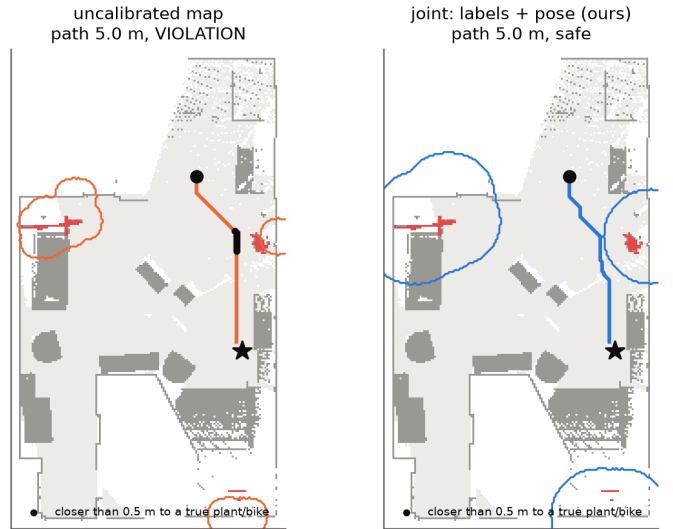


Fig. 1. A hard task at L2 in the planner frame. Red: true plant and bike; line: keep-out boundary. The uncalibrated zone misses the truncated edge of the plant and the path passes within 0.5 m of it (black dots); the joint zone covers it.

TABLE II

MISSIONS WITH PLANT AND BIKE AS AVOID CLASSES: HARD TASKS (PATH FOUND, UNSAFE AMONG FOUND PATHS) AND RANDOM TASKS (SUCCESS).

calibration	hard: path		hard: unsafe		random: success	
	L0	L3	L0	L3	L0	L3
uncalibrated	0.99	0.89	0.55	0.54	0.93	0.80
label sets [5]	0.27	0.21	0.00	0.27	0.43	0.31
geometry only	0.27	0.05	0.00	0.00	0.43	0.09
joint (ours)	0.32	0.30	0.00	0.00	0.78	0.68
oracle	1.00	0.99	0.00	0.00	1.00	0.97

is reachable in 0–3 % of realizations, against 96–100 % for the uncalibrated map. Label sets are safe at the true pose, but 9 and 27 % of their paths are unsafe at 2 and 7 cm of drift. The price is that a path exists in only 30–32 % of these deliberately hard tasks, against 97–100 % on the true map, so the uncalibrated planner has a higher success rate (39–45 % vs. 29–32 %): about half of its paths are safe by luck. The guarantee bounds risk; it does not raise success. Map mIoU does not predict this risk: across 15 scenes its Spearman correlation with the uncalibrated planner’s share of unsafe paths is 0.18 (95 % bootstrap interval –0.35 to 0.68), too few scenes to establish either a relation or its absence (H4). Semantics pays off: the geometry-only arm finds a path in 27 % of the hard tasks at L0 and in 5–15 % under drift, 2–6 times less often than the joint margin. On random tasks the joint margin succeeds in 78 % of the missions at the true pose and in 68 % at 7 cm, with no unsafe path; the uncalibrated planner succeeds in 93 and 80 %, but 7–8 % of its paths are unsafe. Collisions with obstacles, which the guarantee does not cover, reach 3–5 % at 7 cm of drift for the arms whose paths follow walls. With the TV stand as a third avoid class, it falls back to the geometry-only certificate and the joint arm finds a path in only 6–23 % of the hard tasks.

TABLE III
RANDOM TASKS (420 IN 21 SCENES): MISSION SUCCESS. * SOME PATHS ARE UNSAFE.

vocab.	$1 - \alpha$	drift	label sets	geometry	joint	uncal.
closed	0.95	none	0.43	0.43	0.42	0.93*
		L2	0.39*	0.19	0.61	0.89*
		L3	0.31*	0.07	0.50	0.80*
closed	0.90	none	0.43	0.43	0.78	0.93*
		L2	0.39*	0.21	0.74	0.89*
		L3	0.31*	0.10	0.67	0.80*
open	0.95	none	0.43	0.43	0.42	0.93*
		L2	0.39*	0.19	0.46	0.88*
		L3	0.31*	0.07	0.34	0.80*
open	0.90	none	0.43	0.43	0.78	0.93*
		L2	0.39*	0.23	0.74	0.88*
		L3	0.31*	0.09	0.68	0.80*

C. Comparison with label sets

The mission success of [5] comes from a different setting (simulated 70×25 m sites, known pose, five classes, an exploration mode), so we compare the methods on our scenes and random tasks in a 2×2 design: a closed vocabulary (five rows of the detector’s prompt embeddings, equivalent to re-prompting it) or the open one, and $1 - \alpha = 0.95$ (theirs) or 0.9 (ours), each at the true pose and at 2 and 7 cm of drift. Their exploration mode is not reproduced. Label sets give the same result in all four settings (Table III): their threshold is always zero and they keep out every occupied cell, like the geometry-only arm at L0 (0.43 mission success). In their setting (0.95, known pose) the joint margin coincides with them (0.42); with 20 calibration scenes and $\alpha = 0.05$ the plant margin is the largest miss, the plant abstains in 95 % of the scenes and falls back to the same zone. At the true pose the vocabulary does not matter and the risk level makes the difference: at $1 - \alpha = 0.9$ the joint margin succeeds in 78 % of the missions against 43 % with either vocabulary. Under drift at $\alpha = 0.05$ the closed vocabulary helps the joint margin (0.61 and 0.50 vs. 0.46 and 0.34), because the plant abstains less often. Under drift label sets admit unsafe paths (0.4 % of the missions at 2 cm and 1.1 % at 7 cm; a dangerous cell is reachable in 56–60 % of realizations), while the joint margin has none and succeeds more often: 0.46–0.74 at 2 cm and 0.34–0.68 at 7 cm against 0.39 and 0.31. Our absolute success is below the 98.36 % reported by [5] (whose most-likely-label baseline reaches 65.57 %), but our adaptation of their calibration (score over true footprint cells, a 5 cm grid, no exploration mode and no PAC variant) gives 43 % on our scenes, so most of the gap comes from the setting rather than the method; likewise PwC reports 20 % collisions and 53.3 % goals reached on hardware against 66.7 and 33.3 % for its baseline [6], and its geometry-only analogue reaches 43 % on our random tasks against 78 % for the joint margin. We repeated both calibrations at the true pose on grids of 5, 25, 50 and 100 cm: the label-set threshold is zero at every resolution and its keep-out zone takes 64 % of the free space at 5 cm and all of it at 1 m, while in this run

($\alpha = 0.05$, calibration on the other 20 scenes) the joint margin abstains for the plant at every resolution and for the bike from 50 cm. A label set can only say “this cell may be a plant”; a distance says “a plant is nearby”, which stays informative when the detector loses the edges of objects.

D. The price of the guarantee

Raising the risk level barely helps on hard tasks. From $\alpha = 0.1$ to 0.3 the joint margin at L3 shrinks by 0.20–0.25 m and the share of hard tasks with a path goes from 30 to 31 %, while coverage tracks $1 - \alpha$ and the share of realizations with a reachable dangerous cell rises from 3 to 20–27 %. At $\alpha = 0.05$ the margin over 20 scenes is the largest miss, a class falls back in 73 % of realizations, and a path exists in 10 % of the hard tasks. On random tasks at 7 cm the picture is the same: 34 % mission success at $\alpha = 0.05$, 68 % at 0.1 and 71 % at 0.3. The margin is set by the worst calibration scene while $n < 2/\alpha - 1$. With 19 calibration scenes the rank is $n - 1$, and one scene with a missed plant no longer forces abstention: the plant abstains in 0 % of the splits at L0 instead of 63 % and in 11 % at 45 cm instead of 59 %, and the bike margin at 45 cm drops from 1.23 to 1.00 m. Better masks help only in part. MobileSAM [24] prompted with the detector boxes cuts the median bike miss at L0 from 0.43 to 0.05 m, but the worst scene keeps 0.75 m, so the bike margin at L0 does not change; at 45 cm it drops from 1.23 to 1.06 m. The plant margin drops from 0.85 to 0.61 m at L0 but grows from 1.54 to 1.86 m at 45 cm. The region threshold $\lambda_0 = 0.05$ was set on the development scene; $\lambda_0 = 0.02$ gives slightly smaller margins (bike at 45 cm: 1.08 vs. 1.23 m), while 0.1 and 0.2 make the plant abstain in 65–100 % of the splits.

What lowers the price. The plant’s worst miss comes from two small pots in one scene: the main detector takes one for a wall cabinet and does not see the other. YOLO-World v2-x and YOLOE-v8-S [25], with the same 98 prompts, both find the plant: its worst miss drops from 3 to 0.71–0.72 m, plant abstention nearly vanishes and the L0 margin drops from 0.85 to 0.71–0.72 m. But each detector has its own whole-object miss. v2-x confuses the TV stand even more often and marks plants and bikes where there are none (their regions grow by 32 and 10 %), so at 7 cm of drift a path exists in 23 % of the hard tasks instead of 30 %. YOLOE swaps the TV stand in one scene instead of three but loses the bike entirely in two, so the bike abstains in 83–85 % of the splits and a path exists in only 7–16 % of the hard tasks. The bottleneck moves between classes rather than disappearing. The TV stand is swapped because of competitors in the vocabulary: the closed five-class vocabulary certifies it in every split, but the detector then calls benches and beds TV stands and three-class missions get no easier (5–8 % vs. 6–23 %); merging a class with its usual “twins” cannot be done honestly, because on the development scene the TV stand is confused with a screen and a monitor, not with the bench, rug and bed of the test scenes. Growing the class region into adjacent occupied cells by 15–30 cm barely changes the margin (by at most 0.14 m) while the region grows 2–4 times: the worst misses are whole objects, not truncated

edges. Hierarchical conformal calibration [27] over all 10 drift realizations of each calibration scene removes the lottery of drawing one (the spread of the bike margin at 45 cm falls from 0.49 to 0.18 m) but raises its median (1.24 to 1.50 m), and one margin for both classes covers both in 92–94 % of the realizations instead of 88–92 % at a 0.03–0.26 m larger mean margin. A conditional margin would need a feature that predicts the miss; path length does not (Spearman -0.09 to 0.15), and a whole-object miss leaves no trace on the map for any feature to see.

Ensembles and a cascade. Since each detector misses a different class, we fuse the three maps built from the same points, taking per cell the maximum, the mean or the second largest of the class fractions, and report all three rules. The maximum removes abstention for all three classes up to 7 cm and nearly so at 45 cm (plant: 5 %; the TV stand: 0 % instead of 75–95 %), with L0 margins of 0.71, 0.75 and 1.40 m for plant, bike and TV stand and certified coverage 0.92–0.94; class regions grow (at the true pose over all scenes: plant by 55 %, bike by 13 %, TV stand by 39 %). A path then exists in 60 and 47 % of the hard tasks at L0 and L2 (32 and 31 % with one detector) but in 22 % at L3 (30 %): almost all of this difference comes from one scene where the ensemble finds the plant and its zone closes the passage, but random-task success at L3 also drops (64 vs. 68 %), in line with the larger regions. The mean keeps the TV stand abstaining (60–66 %), agreement of two detectors loses it (98–99 %) and solves no more hard tasks than one detector. A cascade that verifies the fused detections with CLIP [26] on box crops (threshold set on the development scene; other thresholds were not tried on test scenes) drops whole plants and TV stands in four test scenes each, and both classes abstain in about 99 % of the splits (a path in 19, 12 and 4 % of the hard tasks); explicit multi-frame voting raises misses already on the development scene. Misses set the margin, so recall matters most and any verification that can only remove detections only adds misses; false positives still cost solvability through larger keep-out zones. The ensemble was proposed after seeing the detectors’ misses on these scenes, so its gain may be optimistic.

Risk instead of abstention. A path with clearance c_k to the region of class k can be unsafe for k only if $s_k > c_k$, so $\rho_k = (1 + \#\{i : s_{i,k} \geq \min(c_k, R_{\max})\}) / (n + 1)$ over the calibration scores is a conformal p-value: $\mathbb{P}(\text{unsafe and } \rho_k \leq a) \leq a$ even for a path chosen on the test map. Planning with a grid of margins and certifying the first path with $\rho_k \leq a/2$ for both classes gives, at $a = 0.2$, a path for 27–29 % of the hard and 70–77 % of the random tasks — the same as the joint margin at $a = 0.1$ — with at most 5.7 % of the certified paths unsafe. It does not raise feasibility, but it replaces abstention by an explicit risk and explains it: with 20 calibration scenes the smallest risk per class is $1/21$, and the scene with the missed plant raises it to $2/21$.

Calibrating missions instead of the map. The joint margin promises that every object of the class lies within it, including objects no path goes near. Conformal risk control [28], already used in planning for control-barrier margins around

humans [29] and in MPC with spectral risk [30] (and close to calibrating the clearance of the chosen trajectory [31]), calibrates what the robot does instead: for a margin m on the grid, the loss $L_i(m)$ of scene i is the fraction of its tasks whose path at margin m is unsafe, made monotone by $\tilde{L}_i(m) = \max_{m' \geq m} L_i(m')$, and the smallest m with $\frac{1}{n+1} \sum_i \tilde{L}_i(m) + \frac{1}{n+1} \leq a$ gives $\mathbb{E}[L_{\text{test}}] \leq a$. This holds for a new scene with at least one task and, like coverage, by construction on leave-one-out splits of a fixed scene set; it is a weaker, mission-level guarantee, and one test scene may be unsafe throughout. Counting success as in the mission tables (no unsafe path, no collision), on random tasks at $a = 0.05$ it succeeds in 94, 86 and 76 % of the missions at L0, L2 and L3, slightly less than the uncalibrated planner (93, 88, 80 %), with at most 0.6 % unsafe missions on average over scenes instead of 7–8 % (the joint margin: 78, 74, 68 %); the worst scene has 13 % unsafe missions. On hard tasks (14 calibration scenes with tasks, so $a = 0.05 < 1/15$ is infeasible) $a = 0.1$ succeeds in 40, 57 and 36 % of the tasks instead of 30–32 %, with up to 8.6 % unsafe missions on average over scenes but one scene unsafe throughout at L0; with the detector ensemble, chosen among several variants on the same scenes, it succeeds in 41, 83 and 52 % with up to 6.7 % on average and the same worst scene.

Execution. The guarantee concerns the plan at query time. When the robot drives each path cell by cell with the motion primitives of the recorded trajectories (3 cm steps, in-place turns of 2°) while its pose estimate keeps drifting at the same level (20 executions per path; without drift an execution is evaluated exactly like the plan), the deviation from the commanded path is 0.3 and 1 % of the distance travelled at L2 and L3 (95th percentile up to 2.9 %), and paths certified at plan time stay safe with respect to the avoid classes: at most 0.6 % of their executions are unsafe, but 1–9 % touch an obstacle, which the guarantee does not cover. At L4 the deviation is about 6 %: up to 2.7 % of the certified paths become unsafe and 20–31 % touch an obstacle. A path can be unsafe during execution only if $s_k + \delta + \varepsilon > c_k$, with δ the largest deviation and $\varepsilon = \sqrt{2} \Delta$ for discretization. Adding $\hat{u}L + \varepsilon$ to the clearance, with $\hat{u}$ a conformal quantile of the deviation per metre pooled over paths of other scenes, restores the guarantee only approximately (paths of a scene are dependent); it is nearly free at L2 (69 vs. 70 % of the random tasks certified), noticeable at L3 (63 vs. 71 %) and leaves 3 % instead of 19 % at L4. Risk control with the loss “unsafe during execution” gives an expected-share guarantee that is exact within the drift model: 84 and 72 % successful random missions at L2 and L3 for $a = 0.05$. At L4 the robot needs relocalization during execution.

E. When does the sum of margins under-cover?

On synthetic drift, which is independent of detector errors, the sum only over-pays, although the premise of Proposition 2 fails in 2–12 % of plant and bike realizations at L1–L4 and up to 26 % at L5. To see the sum under-cover we keep the real label errors and replace drift by a scene-level mixture: in b of

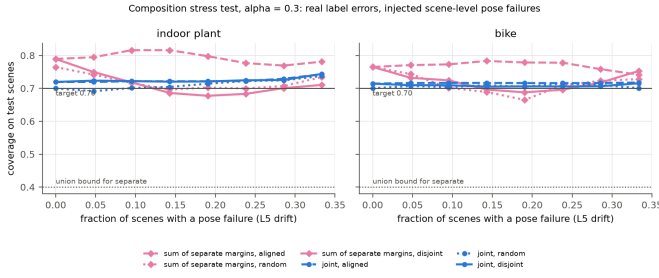


Fig. 2. Coverage of the sum of separate margins and of the joint margin vs. the fraction of scenes with a localization failure placed where labels are worst (dashed), best (solid) or at random (dotted); $1 - \alpha = 0.7$.

21 scenes a localization failure occurs (realizations from L5), in the others drift is negligible (L1). Failures are placed in the scenes with the worst labels, the best labels, or at random; $\alpha = 0.3$, because at $\alpha = 0.1$ with 13 scenes any failure enters the maximum, and 1000 splits. When failures coincide with bad labels the sum over-pays (0.74–0.82); when they hit scenes with good labels it falls to 0.68–0.69 at a 0.70 target (14–24 % failing scenes), and to 0.66 at random. The joint margin gives 0.69–0.74 however failures are placed, close to the rank bound $10/14 \approx 0.71$ (Fig. 2). The under-coverage is small; it shows that the $1 - 2\alpha$ bound is not a formality and that where the sum lands depends on how failures are placed, which is unknown in advance.

F. Real odometry and other scenes

Frame-to-frame odometry without loop closure errs in jumps, mostly during in-place turns near walls: its ATE ranges from 3.5 cm to 3.9 m across scenes, with a median of 78 cm. The labels-only margin then silently covers only 0.62 (bike) and 0.74 (plant). The joint margin is formally valid (0.96), but only because it abstains in 87 % of the splits for both classes; when it certifies, coverage is 0.71–0.72. In missions it finds no path: both classes fall back to the geometry-only certificate, which closes almost all free space at this error. The uncalibrated planner finds a path in 23 % of the hard tasks, and 62 % of those paths are unsafe. With this pose quality there is no useful guarantee. Pose-graph SLAM with loop closures on top of the same odometry, following Choi et al. [32] (keyframes every five frames, ORB-PnP loop candidates verified by RGB-D odometry and point-cloud overlap, thresholds set on the development scene), lowers the median ATE from 78 to 24 cm, below 30 cm in 12 of 21 scenes instead of two. Two scenes find no loop (3.3 and 2.0 m) and set the margin: the labels-only bike coverage is 0.80, the joint margin covers 0.92 at 1.59 m without abstention, but finds a path in only 1 % of the hard tasks; the mission-level guarantee gives 36 % successful random missions at $a = 0.05$ with 0.5 % unsafe, out of 41 % for which the map has any path. The tail of the localization error, not its median, decides. Margins calibrated on ReplicaCAD and tested on HM3D scenes [33] of OSMA-Bench give one counterexample: of the two scenes in which a plant is visible from the trajectory, the detector finds

one (miss 0.18 m) and splits the other across three wrong classes (2.9 m). A milder shift, a held-out layout of the same family (four groups of five scenes and apt_3), keeps the mean joint coverage at 0.94–0.96, but single layouts drop to 0.80 at L0 and 0.63 at 45 cm (bike in the single apt_3 scene).

VII. DISCUSSION AND LIMITATIONS

The joint margin is not cheap: its price is set by the worst calibration scene, and a single object missed by the detector makes its class abstain in a large share of the splits. More calibration scenes weaken this dependence: with 19 of them the plant abstains in 0–11 % of the splits instead of 24–63 %. Three expectations were wrong. Positive correlation between the errors makes the sum of margins more conservative; under-coverage appears when rare failures of the two modules hit different scenes, and it is small. Label sets degenerate because of truncated edges, not complete misses. And real odometry errs in jumps rather than smooth drift. In earlier runs (not repeated) neither convex hulls of the regions nor a confidence-scaled margin shrank the keep-out zone. The bottleneck is confident misses and class swaps of whole objects, which map metrics such as mIoU do not show; a “fraction of scenes with a whole-object miss” metric would fit mapping benchmarks.

Limitations. One family of 21 simulated scenes and one lighting condition; coverage on random splits of these scenes holds by construction, and only bootstrap intervals 0.1–0.15 wide, a held-out layout and two HM3D scenes speak to new scenes. Drift is synthetic and independent of detector errors; real localization is tested with odometry and our own Open3D-based loop-closure SLAM only. The perception is a box detector, not a full mapper. The main guarantee is marginal over calibration sets, per class, and about the plan at query time; execution is tested with the drift model, the execution bound pools paths and is only approximately exchangeable, collisions with obstacles are not covered, and discretization can take up to 7 cm of the safety distance at the true pose and 11 cm under drift. The hard task family and the mission classes were chosen after first runs, and label sets are our re-implementation on final maps, without the authors’ exploration mode.

VIII. CONCLUSION

The miss distance in the planner frame is one calibrated quantity that gives a planner a distribution-free guarantee against label and pose errors together, and abstains when the guarantee would be useless. Margins calibrated on either error alone lose coverage under the other, the sum of separate margins over-pays or under-covers depending on an unknown placement of failures, and label-set calibration degenerates to keeping out everything occupied. The price is set by the worst perception or localization error among the calibration scenes. Fusing detectors for recall and calibrating the share of unsafe missions instead of the map lower it, and execution under drift of up to 7 cm keeps it for the avoid classes; more calibration scenes, SLAM that closes loops in every scene, relocalization

during execution and full open-vocabulary mappers are the next steps.

REFERENCES

- [1] Q. Gu *et al.*, “ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning,” in *Proc. IEEE ICRA*, 2024, pp. 5021–5028, doi: 10.1109/ICRA57147.2024.10610243.
- [2] M. Popov *et al.*, “OSMa-Bench: Evaluating open semantic mapping under varying lighting conditions,” in *Proc. IEEE/RSJ IROS*, 2025, pp. 10786–10791, doi: 10.1109/IROS60139.2025.11247603.
- [3] D. Rotondi *et al.*, “3D scene graphs: Open challenges and future directions,” arXiv:2606.19383, 2026.
- [4] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
- [5] D. S. Sundarsingh *et al.*, “Safe planning in unknown environments using conformalized semantic maps,” *IEEE Robot. Autom. Lett.*, vol. 11, no. 4, pp. 4921–4928, 2026, doi: 10.1109/LRA.2026.3668466.
- [6] Z. Mei *et al.*, “Perceive with confidence: Statistical safety assurances for navigation with learning-based perception,” *Int. J. Robot. Res.*, vol. 45, no. 6, pp. 938–967, 2026, doi: 10.1177/02783649251378151.
- [7] R. Baral, “Marginal calibration does not compose: Hidden dependence in modular robot navigation,” arXiv:2609.23731, 2026 (IROS 2026 workshop).
- [8] R. Kurkova, M. Popov, and S. Kolyubin, “OSMa-Bench++: Toward open-ended benchmarking of semantic mapping for manipulation with prompt-generated synthetic scenes,” arXiv:2605.26831, 2026 (ICRA 2026 workshop).
- [9] J. M. Correia Marques, A. J. Zhai, S. Wang, and K. Hauser, “On the overconfidence problem in semantic 3D mapping,” in *Proc. IEEE ICRA*, 2024, pp. 11095–11102, doi: 10.1109/ICRA57147.2024.10611306.
- [10] V. Tchuiev and V. Indelman, “Epistemic uncertainty aware semantic localization and mapping for inference and belief space planning,” *Artif. Intell.*, vol. 319, p. 103903, 2023, doi: 10.1016/j.artint.2023.103903.
- [11] L. Lindemann *et al.*, “Safe planning in dynamic environments using conformal prediction,” *IEEE Robot. Autom. Lett.*, vol. 8, no. 8, pp. 5116–5123, 2023, doi: 10.1109/LRA.2023.3292071.
- [12] S. Yang, G. J. Pappas, R. Mangharam, and L. Lindemann, “Safe perception-based control under stochastic sensor uncertainty using conformal prediction,” in *Proc. IEEE CDC*, 2023, pp. 6072–6078, doi: 10.1109/CDC49753.2023.10384075.
- [13] J. Shin, Y. Ra, and I. Yang, “From prediction uncertainty to conformalized distance fields for safe motion planning,” arXiv:2607.00776, 2026.
- [14] E. M. Kim and J.-K. Kim, “SCORE: Shape-conforming regions for flight in enclosed, degraded environments,” arXiv:2608.15289, 2026.
- [15] D. Kumar *et al.*, “Learnable conformal prediction for safe and efficient robotics under perception and planning uncertainties,” in *Proc. IEEE ICRA*, 2026, pp. 14129–14136, doi: 10.1109/ICRA57385.2026.11697249.
- [16] S. Natraj, B. Sinopoli, and Y. Kantaros, “Conformal constraint tightening for chance-constrained motion planning with unknown dynamics,” arXiv:2607.22409, 2026.
- [17] G. C. Calafiore, “Bridging conformal prediction and scenario optimization: Discarded constraints and modular risk allocation,” arXiv:2603.19396, 2026.
- [18] R. Mahajan, A. Raghavan, and K. H. Johansson, “Finite-sample conformal coverage recovery via fusion under degraded local guarantees in occupancy map estimation,” arXiv:2607.14906, 2026.
- [19] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, “Conformal prediction beyond exchangeability,” *Ann. Statist.*, vol. 51, no. 2, 2023, doi: 10.1214/23-AOS2276.
- [20] H. Sier *et al.*, “P-POSEMEM: Projective semantic memory for consistent language grounding under pose-graph rewrites,” arXiv:2609.15475, 2026.
- [21] A. Szot *et al.*, “Habitat 2.0: Training home assistants to rearrange their habitat,” in *Proc. NeurIPS*, 2021, arXiv:2106.14405.
- [22] T. Cheng *et al.*, “YOLO-World: Real-time open-vocabulary object detection,” in *Proc. IEEE/CVF CVPR*, 2024, pp. 16901–16911, doi: 10.1109/CVPR52733.2024.01599.
- [23] Q.-Y. Zhou, J. Park, and V. Koltun, “Open3D: A modern library for 3D data processing,” arXiv:1801.09847, 2018.
- [24] C. Zhang *et al.*, “Faster segment anything: Towards lightweight SAM for mobile applications,” arXiv:2306.14289, 2023.
- [25] A. Wang *et al.*, “YOLOE: Real-time seeing anything,” in *Proc. IEEE/CVF ICCV*, 2025, pp. 24591–24602, doi: 10.1109/ICCV51701.2025.02280.
- [26] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, PMLR vol. 139, 2021, pp. 8748–8763, arXiv:2103.00020.
- [27] Y. Lee, R. F. Barber, and R. Willett, “Distribution-free inference with hierarchical data,” *ACM J. Data Sci.*, 2026, doi: 10.1145/3786352.
- [28] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, “Conformal risk control,” in *Proc. ICLR*, 2024, arXiv:2208.02814.
- [29] J. Gonzales, K. Mizuta, K. Leung, and L. J. Ratliff, “Safe probabilistic planning for human-robot interaction using conformal risk control,” in *Proc. IEEE/RSJ IROS*, 2025, pp. 18676–18683, doi: 10.1109/IROS60139.2025.11247339.
- [30] J. Eom and T. Ersal, “Distribution-free risk-aware planning and control under uncertainty using conformal spectral risk control,” arXiv:2606.04185, 2026.
- [31] P. Y. Chang and Q. Ahmed, “Barrier function conformal safety clearance certification with CVaR for driving trajectory selection,” arXiv:2608.26533, 2026.
- [32] S. Choi, Q.-Y. Zhou, and V. Koltun, “Robust reconstruction of indoor scenes,” in *Proc. IEEE CVPR*, 2015, pp. 5556–5565, doi: 10.1109/CVPR.2015.7299195.
- [33] S. K. Ramakrishnan *et al.*, “Habitat-Matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI,” in *NeurIPS Datasets and Benchmarks*, 2021, arXiv:2109.08238.